TARFORM

Summary of prior work

Tarform Motorcycles has two, stacked Printed Circuit Board Assemblies (PCBA) that function together as a motorcycle's Bike Control Unit (BCU). The first PCBA, the "Carrier Board" (CB), is an 8 layer, low power circuit holding an edge card connectorized System On Module, multiple LGA radio modules, and automotive connectors. The CB exchanges low speed digital signals with the "Secondary Board" (SB), a 2 layer, 1 sided PCBA with sections of isolated, high voltage ground, and a single automotive connector. Design for both boards was done with Cadence OrCAD + Allegro v17.4 software.

Statement of work

The Secondary Board requires trivial updates to eliminate rework. The Carrier Board needs a major engineering effort to redesign several peripheral systems. Tarform will be working in tandem for support. All designs must meet applicable automotive standards.

Feel free to redesign and move things around if needed. The End Connectors need to be in the same place. ie HDMI, USB, power, pin connector.

The Carrier Board is for carrying the Toradex Apalis iMX8
https://www.toradex.com/computer-on-modules/apalis-arm-family/nxp-imx-8

Deliverables

- (Optional) Export of the ECAD in Cadence 16.2 file format.
- Conversion of both boards' ECAD into either Altium Designer or KiCad.

On the secondary board,

- redesign a single transistor relay driver circuit, and implement changes in BOM and PCB layout for two, low power, low speed signals.
    - Analyze the existing circuit for its design flaws.
    - Verify the base/gate drive requirements and resistor values.
    - Ensure that power ratings and layout considerations are followed.
    - Trace the paths of the two specific low-power, low-speed signals.
    - Redesign as required, ensuring optimal trace lengths and minimal interference with other signals.
- add parallel alternate stacking connector, current one is not secure enough with loose connections.
- add power indication LED

On the carrier board,

- Replace an existing board mounted LGA LTE modem with an M2 connector for use with a M.2 edge card LTE module.
    - Ensure impedance-controlled routing for RF and high-speed digital lines.
    - Identify space on the board for the M2 connector.
- Modify the existing SD-Card circuit to conform to a vendor provided reference design.
    - Theres a reference design on the "Ixora v1.2" carrier board from Toradex that needs to be followed for the SD card.
    - https://www.toradex.com/products/carrier-board/ixora-carrier-board
- Remove the board mounted LGA WiFi+Bluetooth module and associated circuits
    - Ensure no traces or components are left floating.
- add 2 line-level audio outputs and 1 microphone input to filtered audio circuit and audio connectors conforming to a vendor provided reference design.
- replace several under-rated power filter passives.
    - Identify the under-rated components.
    - Choose replacements with higher power ratings, ensuring that footprints are compatible or update the layout if needed.
- CAN Bus Channels:
    - Debug the non-working CAN Bus channels. There should be 3. Only one is working. Check for software or hardware issues.
    - Ensure termination resistors are appropriately placed and of the correct value.
- Connect R232 chip VCC to 3.3V
- add parallel alternate stacking connector
- add power indication LED
- add a two pin header that triggers module recovery mode

Simulation & Verification:

- Once the design modifications are done, simulate the circuits for functionality.
- Use time-domain and frequency-domain analyses for the high-speed components, ensuring signal integrity, crosstalk avoidance, and compliance with EMC/EMI standards.

- Cross-referencing is crucial to ensure that no detail is overlooked and all changes are consistent.